

Cell Injury, Adaptation and Cell Death

Pathology — Lecture 3

Dr. Sarah Mahmoud

Faculty of Medicine, 2025–
2026

Learning objectives

- By the end of this lecture you should be able to:
 - List the causes of cell injury
 - Describe the morphology of reversible and irreversible injury
 - Compare necrosis and apoptosis

Causes of cell injury

- Oxygen deprivation (hypoxia) is the most common cause of cell injury.
- Physical agents: trauma, extremes of temperature, radiation and electric shock
- Chemical agents and drugs
- Infectious agents: viruses, bacteria, fungi and parasites
- Immunologic reactions
- Genetic derangements
- Nutritional imbalances

Hypoxia versus ischemia

Hypoxia is a deficiency of oxygen that causes cell injury by reducing aerobic oxidative respiration.

Ischemia is a loss of blood supply to a tissue. It impairs the delivery of substrates for glycolysis as well as oxygen. Therefore, ischemia injures tissues faster than hypoxia alone.

Feature	Hypoxia	Ischemia
Oxygen delivery	Reduced	Reduced
Anaerobic glycolysis	Continues	Stops
Speed of injury	Slower	Faster

Reversible cell injury

The two main morphologic correlates of reversible cell injury are cellular swelling and fatty change.

Cellular swelling is the first manifestation of almost all forms of injury to cells. It results from failure of energy-dependent ion pumps in the plasma membrane.

Fatty change is seen mainly in cells involved in lipid metabolism, such as hepatocytes and myocardial cells. It is manifested by the appearance of lipid vacuoles in the cytoplasm.

Mechanisms of cell injury

- ATP depletion causes failure of the Na⁺/K⁺ ATPase pump, leading to influx of sodium and water and cellular swelling.
- Decreased ATP causes a switch to anaerobic glycolysis, which leads to lactic acid accumulation and decreased intracellular pH.
- Influx of calcium activates phospholipases, proteases, endonucleases and ATPases.
- Mitochondrial damage leads to formation of the mitochondrial permeability transition pore.
- Oxygen-derived free radicals cause lipid peroxidation of membranes.

Think about it

Why does ischemia cause more rapid injury than hypoxia?

What happens to the cell if the insult persists?

Irreversible injury

Two phenomena consistently characterize irreversibility: the inability to restore mitochondrial function and profound disturbances in membrane function.

There are two patterns of cell death: necrosis and apoptosis.

Nuclear changes in necrosis

Pyknosis: nuclear shrinkage and increased basophilia

Karyorrhexis: fragmentation of the pyknotic nucleus

Karyolysis: fading of basophilia due to DNase activity

Patterns of tissue necrosis

Pattern	Typical location	Key feature
Coagulative necrosis	Infarcts of heart and kidney	Preserved tissue architecture
Liquefactive necrosis	Brain infarcts and abscesses	Digestion of dead cells into a liquid mass
Caseous necrosis	Tuberculosis	Friable, cheese-like white material
Fat necrosis	Pancreas and breast	Chalky white calcium soap deposits
Fibrinoid necrosis	Blood vessel walls	Deposition of immune complexes and fibrin

Gangrenous necrosis

Dry gangrene is coagulative necrosis of a limb that has lost its blood supply. Wet gangrene occurs when bacterial infection is superimposed, producing liquefactive necrosis.

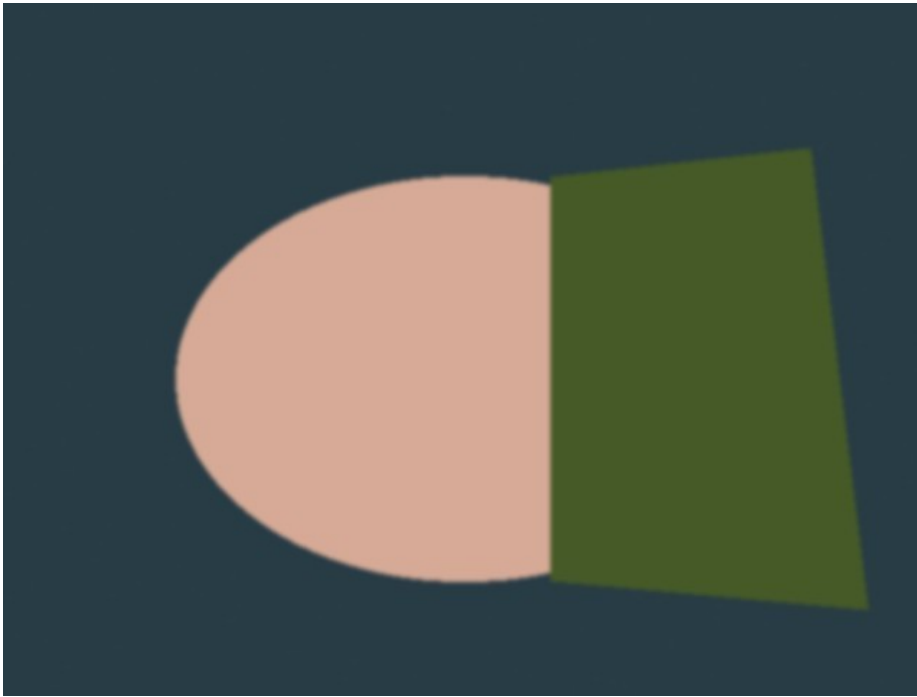


Figure 1: Wet gangrene of the foot

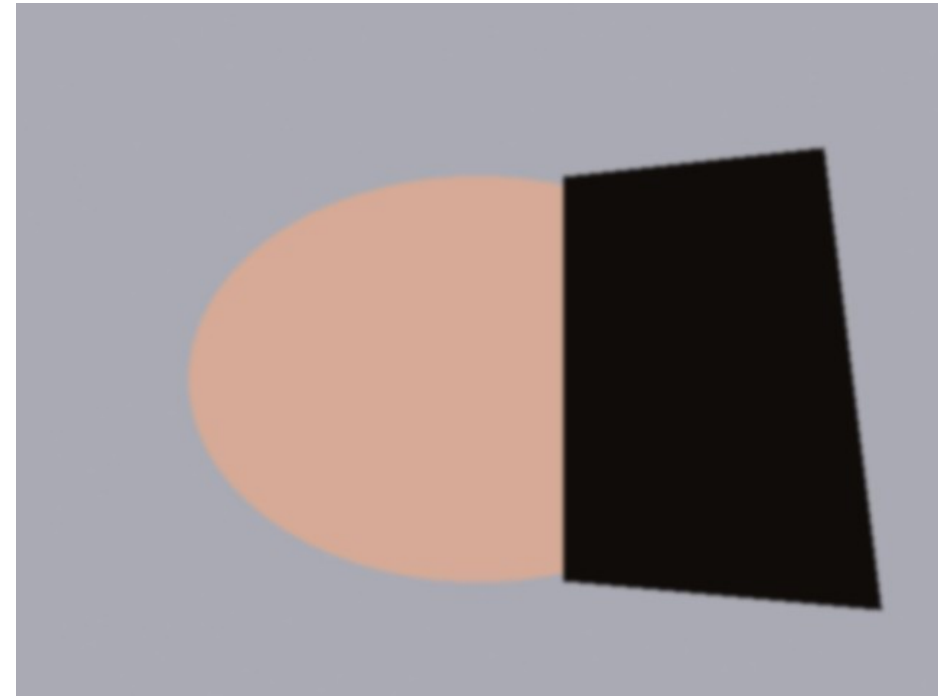


Figure 2: Dry gangrene of the toes

Apoptosis: two pathways

Intrinsic (mitochondrial) pathway

The intrinsic pathway is triggered by loss of survival signals, DNA damage and misfolded proteins. It is regulated by the BCL-2 family of proteins. Release of cytochrome c from mitochondria activates caspase-9.

Extrinsic (death receptor) pathway

The extrinsic pathway is initiated by engagement of death receptors such as Fas and the TNF receptor. It activates caspase-8. Both pathways converge on executioner caspases 3 and 6.

Necrosis versus apoptosis

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Plasma membrane	Disrupted	Intact
Adjacent inflammation	Frequent	None
Physiologic role	Invariably pathologic	Often physiologic

Intracellular accumulations and calcification

- Lipofuscin is a yellow-brown “wear-and-tear” pigment associated with aging.
- Hemosiderin is a hemoglobin-derived pigment that accumulates in tissues with iron excess.
- Dystrophic calcification occurs in dead or dying tissues despite normal serum calcium levels.
- Metastatic calcification occurs in normal tissues whenever there is hypercalcemia.

References

- Kumar V, Abbas AK, Aster JC. Robbins and Cotran Pathologic Basis of Disease. 10th ed. Elsevier; 2020.
- Lecture slides are for educational use only.

Thank you

Questions?